

Lesson 42: Definite Integrals & Area — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Evaluate $\int_0^2 2x \, dx$.
2. Evaluate $\int_1^2 3x^2 \, dx$.
3. Evaluate $\int_0^1 4x \, dx$.
4. Do you add + C for a definite integral?
5. Evaluate $\int_0^3 2x \, dx$.
6. Evaluate $\int_1^2 2x \, dx$.

Section 2 — Core practice

7. Evaluate $\int_1^3 (2x + 1) \, dx$.
8. Evaluate $\int_0^2 (3x^2) \, dx$.
9. Evaluate $\int_1^4 4x \, dx$.
10. Evaluate $\int_0^2 6x^2 \, dx$.
11. Evaluate $\int_1^3 x^2 \, dx$.

12. Evaluate $\int_0^1 (6x^2 + 2) \, dx$.

Section 3 — Challenge & exam-style

13. Find the area between $y = x^2$, the x-axis, $x = 0$ and $x = 3$.

14. Evaluate $\int_1^4 (2x + 1) \, dx$ and state the area under $y = 2x+1$ over that interval.

15. Find the area between $y = x^2$ and the x-axis from $x = 1$ to $x = 2$.

16. Evaluate $\int_0^2 (4x^3) \, dx$ and interpret as an area.